

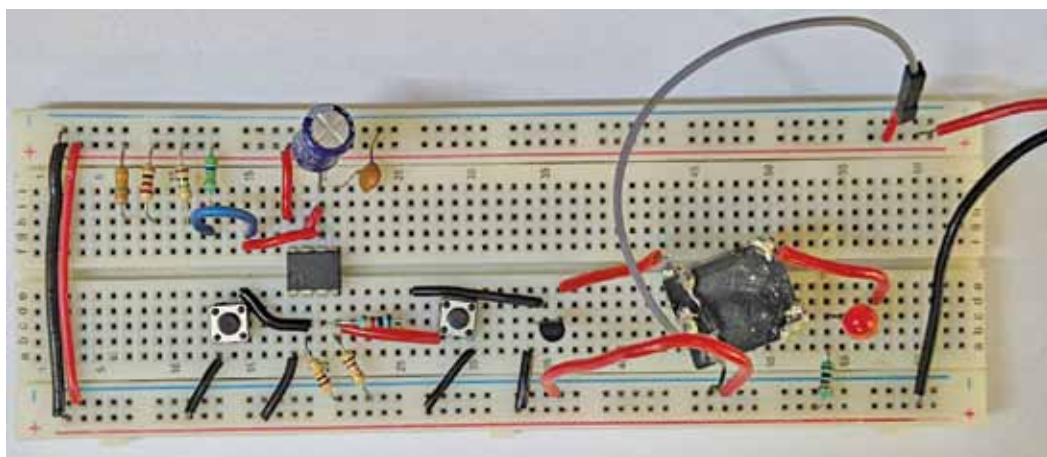
**T**his delay-off timer is a cost-effective, reliable, and easy-to-assemble device for automating time-based switching operations. It improves energy efficiency and adds convenience in both home and office settings by ensuring appliances remain on only as long as required. Delay-off timers are commonly used in appliances that need to remain active for a defined duration even after the control switch or trigger has been turned off. They contribute

Typical uses include exhaust fans in bathrooms and kitchens, where the timer keeps the fan running briefly to eliminate residual moisture or odours. In staircases or corridors, it ensures lights remain on just long enough for safe passage. Other applications include automatic water pumps, car interior lighting, and buzzers or alarms.

a TLC555 IC configured in monostable mode. Once the start switch is pressed, the circuit powers the connected appliance for a preset interval before automatically switching it off. Dedicated push switches allow selection of delay options—5, 10, 15, 20, or 30 minutes—based on the switch used. Fig. 1 shows the timer circuit on a breadboard.

Fig. 2 shows the circuit diagram of the delay-off appliance timer. It is

powered by 230V AC mains supply, which is stepped down to 12V, 500mA by a transformer, rectified using a bridge rectifier (BR1), filtered by capacitor C1 (1000 $\mu$ F, 35V), and regulated to 12V using an LM7812 (IC1). This regulated 12V supply powers the remaining circuit.



The diagram illustrates a power supply circuit for a 230V AC to 12V AC transformer. The circuit components and their connections are as follows:

- Transformer (X1):** A 230V AC primary transformer with a 12V AC secondary transformer (500mA).
- Rectifier (BR1):** A bridge rectifier (1A) connected to the secondary of X1.
- Voltage Regulator (IC1 LM7812):** A 12V regulator connected to the rectifier output. It includes a 1000uF, 35V capacitor (C1) and a 1K resistor (R1).
- 555 Timer (IC2 TLC555):** A timer configured as a square wave generator. It includes a 580K resistor (R4), a 1.2M resistor (R5), a 1.8M resistor (R6), a 2.2M resistor (R7), a 3.3M resistor (R8), a 500K resistor (R9), a 470uF, 35V capacitor (C2), and a 10nF capacitor (C3).
- Relay (RL1 SPDT RELAY):** A single-pole double-throw relay controlled by the 555 timer output. It is connected to a 230V AC source (CON2) and a 12V AC source (CON3).
- Diodes (D1, D2):** 1N4007 diodes used for rectification and protection.
- Transistors (T1, T2):** 2N2219 transistors used for switching.
- Switches (S1, S2, S3, S4, S5, S6, S7):** Various switches for controlling the circuit.
- LEDs (LED1, LED2):** LEDs for status indication.

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